

Evaluating Subjective, Nondeterministic Agent Outputs

A defensible evaluation plan for a **Deep Search agent** — an agent that plans, browses, reads, and synthesizes a cited research answer — built to survive an adversarial round. Naming *LLM-as-judge* or *user feedback* is the opening move, not the answer.

OUTPUT IS	THE TRAP	THE DISCIPLINE
Subjective + Nondeterministic	Confidence $\neq$ correctness	Signal · Source · Failure mode

§00 Say this first — the framing

“Before I name any evaluator, I want to separate three things that usually get conflated: the **quality dimension** I'm trying to measure, the **source** I get a signal from, and that source's **failure mode**. Subjectivity and nondeterminism break measurement in different ways, so I'll treat them separately — and I'll trust **no single source as ground truth** unless it's anchored to something more trustworthy.”

Everything below is one elaboration of that sentence. The recurring move is to push as much of “quality” as possible *out* of the subjective bucket into things I can check, then handle the irreducibly subjective remainder with calibrated, cross-checked instruments.

§01 Two words, two different problems

“Subjective” and “nondeterministic” are not the same difficulty. Conflating them is the first mistake; name which one each mitigation addresses.

NONDETERMINISTIC Same input → different output Different search paths, different sources surfaced, different synthesis each run. Attacks the reliability of the measurement — a single run tells you almost nothing. ----- FIX → evaluate distributions, not points. Measure variance as its own metric. Fix seeds for regression.	SUBJECTIVE No single ground-truth label “Is this a good answer to an open research question?” depends on rater, intent, and context. Attacks the existence of a label to grade against. ----- FIX → decompose into checkable sub-criteria. Multiple raters + measured agreement. Reference-free rubrics.
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§02 The lens I never drop — signal / source / failure mode

Every evaluator in the plan gets described as three separable things. When the interviewer attacks one, I answer on that axis without conceding the others.

01 · QUALITY SIGNAL <i>The dimension of goodness — faithfulness, coverage, relevance, usefulness. Exists independent of who measures it.</i>	02 · SOURCE <i>Who / what produces the estimate — code check, LLM judge, human expert, a user's click. Cheap-and-scalable trades against trustworthy.</i>	03 · FAILURE MODE <i>How the source lies — bias, miscalibration, sparsity, confounds. Every source has one; the plan names it and guards it.</i>
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Rule that falls out of the lens: **a source can only be trusted as far as its anchor**. A cheap source is validated against a more trustworthy one, forming a chain that terminates in human judgment and, ultimately, real user outcomes.

§03 Decompose “good” into a rubric — move quality out of the subjective bucket

The single most important move: stop grading one holistic “quality” score. Break the Deep Search output into dimensions, and for each one ask whether it is **objectively checkable** or **requires judgment**. The checkable ones stop being subjective at all.

DIMENSION	TYPE	HOW I'D MEASURE IT	BEST SOURCE
Citation validity	CHECKABLE	URL resolves; cited quote actually appears in the source; no fabricated links.	Deterministic code
Faithfulness / grounding	SEMI	Every claim traceable to a retrieved source; no unsupported assertions (NLI / entailment against sources).	Code + LLM judge
Key-fact coverage	SEMI	Recall of the must-hit facts/sources for that query (needs a curated gold set).	Gold set + human
Recency / freshness	CHECKABLE	Source dates vs. query's time-sensitivity; stale-source flag.	Deterministic code
Relevance to intent	JUDGMENT	Did it answer what was actually asked, incl. implicit intent? Pairwise vs. baseline.	Judge + user signals
Comprehensiveness	JUDGMENT	Balanced across viewpoints; no major gap or one-sidedness.	Human + judge
Synthesis / usefulness	JUDGMENT	Insight beyond a link dump; actionable, well-structured. Revealed by behavior.	User signals + human
Calibrated hedging	JUDGMENT	States uncertainty where sources conflict; doesn't overclaim.	Human review

Why this matters against every later objection: the **CHECKABLE** rows need no judge and no user — a confidently-wrong citation is caught by code regardless of how good the prose is. That's the correctness floor no preference signal will ever protect.

§04 The evaluation stack — trust traded for scale

Five layers. The top is trustworthy but tiny; the bottom is huge but noisy. Each layer is **calibrated by the one above it** — that anchor chain is the whole design.

L0	Deterministic / programmatic checks Citations resolve, quotes match, format valid, latency & cost budget, task completed. Runs on <i>every</i> output.	TRUST ●●●● / SCALE ●●●● coverage narrow
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L1	Curated gold set + human expert labels Research questions with known must-hit facts and reference answers. Small N, expensive — the calibration anchor for everything below.	TRUST ●●●● / SCALE ●○○○ the ground truth
L2	LLM-as-judge, calibrated against L1 Scores subjective dimensions offline at scale. Trusted <i>only</i> to the degree its agreement with L1 is measured. See §06.	TRUST ●●●○ / SCALE ●●●○ needs calibration
L3	Actively-sampled human review in production Stratified sampling (by query type, judge disagreement, low confidence, high impact). Keeps L2 honest and catches drift.	TRUST ●●●○ / SCALE ●●○○ we control the frame
L4	Online behavioral signals + A/B Real usage at full volume. Measures <i>usefulness/relevance by revealed preference</i> — and is blind to correctness . See §09–§10.	TRUST ●○○○ / SCALE ●●●● relevance, not truth

The anchor chain (say it): human labels calibrate the judge → the judge scales offline evaluation → online experiments confirm the offline gains are real → programmatic checks guard the correctness that no preference signal protects.

§05 Handling nondeterminism head-on

01 · EVALUATE DISTRIBUTIONS Run each eval input k times ; report mean, variance, and <i>worst-case</i> , not a single pass. A metric on one run is noise.	02 · VARIANCE IS A METRIC A research agent that returns wildly different answers to the <i>same</i> query is worse. Track cross-run consistency as a first-class score.
03 · PREFER RANKING Pairwise / relative judgments are far more stable across runs than absolute scores. Grade “A vs B,” not “7/10.”	04 · SEED FOR REGRESSION Fix seed/temperature on the regression suite so a metric diff is attributable to a <i>code change</i> , not sampling. Keep a separate sampled suite for real-world variance.

§06 “Your LLM judge is overconfident.”

Concede the right thing, immediately: “Agreed — and I'd separate the judge's *confidence* from its *correctness*. An LLM emits a fluent 9/10 whether or not it's right; the stated confidence carries no information about calibration. So I never consume the judge's score as truth — I treat it as a *prediction of the human label with a measured error rate*.”

- 01 **Calibrate against humans; report agreement, not confidence.** Measure judge↔human agreement (Cohen's κ / Krippendorff's α) on a held-out set. If $\kappa = 0.4$, it's a weak instrument and I weight it accordingly. The number I publish is the judge's *accuracy vs. L1*, never its self-rated certainty.
- 02 **Prefer pairwise over absolute.** “Which answer is better, A or B?” is lower-variance and more robust than Likert grading — and sidesteps the fake precision of a 0–10 scale.
- 03 **Force rubric-bound, grounded scoring.** The judge must cite *which criterion* and *which span* justifies the score. This converts vibes into auditable claims a human can spot-check, and confines the judge to what it can actually verify.
- 04 **Ensemble + treat disagreement as signal.** Multiple models/prompts; where they disagree, route to human review. Disagreement flags genuine ambiguity — it isn't noise to be averaged away.
- 05 **Perturbation / trap tests.** Swap a valid citation for a fabricated one — does the score drop? If not, the judge is blind to the dimension I care about. Test the judge like any other model.
- 06 **Guard the known biases.** Randomize order (position bias), control for length (verbosity bias), don't judge with the generator's own model family (self-preference), watch for sycophancy. Judge only what it's good at — *faithfulness of text to provided text* — not “is this the objectively best research answer.”

Land it: “The judge is a cheap, scalable, biased instrument. It's useful exactly to the extent I've measured its error against a trusted anchor — and useless the moment I stop.”

◆ OBJECTION 2 — DEFEND THIS

§07 “Users won't give you explicit feedback.”

Concede + reframe: “Right — thumbs rates are ~1–5% and *non-random*: the delighted and the furious click; the silent middle doesn't. So explicit feedback is sparse *and* selection-biased. I'd never make it the primary metric — it's one low-weight, high-bias source. The fix isn't to beg harder for feedback; it's to stop depending on volunteered labels.”

- 01 **Actively sample instead of waiting.** I control the sampling frame for L3 human review, so I stratify deliberately — by query type, judge disagreement, low confidence, high impact — instead of accepting whoever self-selects. This is the real substitute for missing feedback.
- 02 **Instrument implicit behavior.** Behavioral signals are emitted by *every* session, not the vocal 3%. They cover the silent middle explicit feedback misses (→ §09).
- 03 **De-bias the feedback I do get.** Treat it as a biased sample: segment by cohort, and read *changes over time within a cohort* rather than absolute levels — the bias is more stable than the level.
- 04 **Lower the friction, raise the structure.** Ask at the moment of value, and make it diagnostic (“which part was wrong — a source? a claim? the coverage?”) so the rare signal is actionable, not just a mood.

◆ OBJECTION 3 — THE DEEPEST ONE

§08 “Your criteria just encode developer bias.”

Concede fully — this is real: “The rubric is a *hypothesis* about what ‘good’ means, written by people who aren't the users and who are motivated to pass their own agent. I can't eliminate that bias. What I can do is make it **explicit, auditable, and continuously falsified** against something the developers don't control — real user outcomes.”

THE FAILURE MODES I'M GUARDING

Measure-what's-easy Rubric drifts toward checkable proxies and drops what actually matters but is hard to score.	Wrong mental model Criteria reflect the developer's idea of the task, not the user's real goal.
Goodhart / teach-to-test The agent overfits the rubric; the score rises while real quality doesn't.	Blind spots A whole dimension users care about is simply absent from the rubric.

- 01 **Derive criteria from user evidence, not a whiteboard.** Ground the rubric in real queries, support tickets, and analyses of actual failures. It's a living artifact revised against production data — each criterion records *why* it exists and what user evidence backs it.

- 02 **Separate who authors from who validates.** Blind external raters — ideally domain experts and real users — check that rubric score *correlates with actual user satisfaction / task success*. If rubric goes up while observed success doesn't, the rubric is wrong, not the users.
- 03 **Hold out an ungameable outcome metric.** Behavioral task success — did the user get their answer and stop searching / act on it — measured independently of the rubric. The rubric *predicts*; the outcome *validates*. Goodhart's law is defeated only by a metric the optimizer can't see.
- 04 **Red-team the rubric.** Ask: “how could an output score perfectly and still be useless?” Every hole found is a missing criterion or a bad proxy. Diverse authorship widens the set of blind spots caught.

§09 Can online signals substitute for missing feedback?

The thesis: “No — they're not a substitute, because they measure a *different thing*. But they partially close the gap, and for the *usefulness* axis they're often **better** than explicit feedback: they're *revealed* preference from every session, not *stated* preference from a biased 3%. What they can never establish is **correctness**.”

Each signal below is stated as three things: what behavior it is, what it can *prove*, and the confound that limits it. Note the asymmetry — most signals are ambiguous, and **absence of a signal is never itself a signal**.

SIGNAL	WHAT IT CAN PROVE	WHAT IT CAN'T — THE CONFOUND
Copy / export / save	Strong positive: valued enough to extract and keep.	Not correctness — user may be copying to critique or fact-check. Confidently-wrong prose gets saved too.
Regenerate / “try again”	Clear negative on <i>this</i> sample; per-query regen rate is a direct nondeterminism-quality metric.	Doesn't say <i>why</i> it was rejected — wrong, slow, or just curiosity about variance.
Follow-up / reformulation	Reformulating the <i>same intent</i> → the answer didn't satisfy.	Natural deepening of a research thread looks identical — could be a satisfied user going further.
Citation click-through	Engagement and trust-checking — the user is verifying.	Ambiguous: <i>high</i> clicks can mean the answer was insufficient and they had to read sources themselves.
Dwell time	Read vs. skimmed.	Very long = engaged <i>or</i> confused; very short = instant-satisfy <i>or</i> dismissal. Confounded by length.
Task resumption / return	Returning to the same task later → it was left unfinished.	Attribution: they may return for a genuinely new, related need.
Session ends on our answer	The gold outcome for search: no further searching elsewhere → need met.	Only observable if we own the whole session; “gave up” ends the session too.
Downstream action	Strongest: used the answer in a doc, cited it, made a decision.	Rarest and hardest to attribute; usually outside our instrumentation.
Abandonment / silence	<i>Nothing reliable.</i>	The core trap: satisfied-and-done and gave-up look identical. Absence of signal ≠ signal.

§10 What online signals establish — and the correctness blind spot

THEY DO ESTABLISH Population-level relevance and usefulness by revealed preference — “did users act as if this was useful?” — cheaply, continuously, and across the silent majority. Great for the subjective/usefulness axis and for detecting regressions in the aggregate.	THEY CANNOT ESTABLISH Correctness / factual accuracy, and <i>why</i> something failed. A beautiful, confidently-wrong Deep Search report gets copied and shared just fine — engagement rewards it. For a research agent that's the most dangerous failure, and behavioral signals are silent on it.
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The one-liner: online signals measure “*did the user behave as if this was useful,*” which is a proxy for **relevance**, not a measure of **truth**. So they close the gap on the subjective axis while the verifiable-correctness axis stays with L0 citation checks + L1/L3 expert review. Never let a preference signal stand in for a correctness signal.

TURNING CORRELATION INTO TRUST

Raw online metrics are correlational and riddled with confounds — selection bias, UI/placement effects (a bigger button gets more clicks), a latency confound (slow answers are abandoned regardless of quality), and novelty effects. To make them trustworthy you need the counterfactual: **A/B tests or interleaving**, where a deliberate change to the agent produces a change in the behavioral metric. That's the discipline — *offline evals predict, online experiments confirm*. A metric you can't move on purpose isn't yet evidence of quality.

§11 Cost & latency — quality has a budget

Why it belongs in the eval, not the SRE dashboard: “For a Deep Search agent, cost and latency *are* quality dimensions, not side constraints. The output isn't one token stream — it's a multi-step plan → search → browse → read → synthesize loop, and each step spends money, wall-clock, and the user's patience. A marginally better answer that takes 4× longer and 5× the tokens is often the *worse* product. So I evaluate quality-per-unit-cost, not quality alone.”

WHAT I MEASURE — CHEAP, DETERMINISTIC, PER-RUN

End-to-end latency Time to first token <i>and</i> time to complete answer. Report the distribution (p50 / p95 / p99), never the mean — the tail is where users rage-quit.	Total cost per query Token spend + tool/search API calls + retrieval. Attribute per step so a runaway loop is visible, not just the total.
Step / tool-call count How many search/read iterations to converge. Rising step count at flat quality = the agent is thrashing, not thinking.	Redundant / wasted work Duplicate fetches, re-reading the same source, tokens spent on sources that never make the answer.

- 01 **Evaluate the frontier, not a single point.** Plot quality vs. cost (and quality vs. latency) as a Pareto curve. A change is a win only if it moves the frontier — better quality at equal cost, or equal quality cheaper. This kills “we improved quality” claims that just bought it with 3× spend.
- 02 **Set budgets as hard gates in the regression suite.** p95 latency and cost-per-query ceilings that fail CI, so a quality tweak can't silently regress spend. Deterministic — these sit in L0 and run on every output.
- 03 **Segment by query difficulty.** A hard multi-hop research question *should* cost more than a lookup. Averaging them hides both waste on easy queries and starvation on hard ones — so budget and grade per difficulty tier.
- 04 **Watch latency's confound in online metrics.** Ties back to §10 — slow answers are abandoned *regardless* of quality, so latency contaminates every behavioral signal. I control for it (or hold it fixed in an A/B) before reading engagement as a quality verdict.

Where it sits in the plan: cost/latency are the most *trustworthy* metrics I have — objective, deterministic, no judge or user required — so they anchor the bottom of the stack alongside L0. The discipline is refusing to read any quality number without its cost and latency printed next to it.

§12 The three sentences to land

- 1 “I keep the **quality signal, its source, and its failure mode** separate — so an attack on one evaluator never collapses the whole plan.”
- 2 “I trust **nothing as ground truth without an anchor** — human labels calibrate the judge, the judge scales, experiments confirm, and code guards the correctness floor.”
- 3 “I never let a **preference signal substitute for a correctness signal** — usefulness and truth are different axes, and a Deep Search agent has to win both.”